

ITA0605 - Machine Learning

List of Lab Exercises

1. Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim:

To implement and demonstrate the **FIND-S algorithm** for finding the most specific hypothesis from a given set of training data.

Algorithm:

1. Read the training dataset.
2. Initialize the hypothesis with the first positive example.
3. Compare the hypothesis with each positive training example.
4. If an attribute differs, replace it with ?.
5. Ignore all negative examples.
6. Display the final most specific hypothesis.

Dataset 1:

Program:

```
data = [  
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],  
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],  
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],  
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']  
]  
  
hypothesis = data[0][: -1]  
  
for row in data:  
    if row[-1] == "Yes":  
        for i in range(len(hypothesis)):  
            if hypothesis[i] != row[i]:  
                hypothesis[i] = "?"  
  
print("Final Hypothesis:", hypothesis)
```

Output:

```
data = [
    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],
    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],
    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']
]

hypothesis = data[0][:-1]

for row in data:
    if row[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] != row[i]:
                hypothesis[i] = "?"

print("Final Hypothesis:", hypothesis)

... Final Hypothesis: ['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

Dataset 2:

Program:

```
data = [
    ['High', 'Good', 'Permanent', 'Yes', 'Young', 'Yes'],
    ['High', 'Good', 'Permanent', 'No', 'Middle', 'Yes'],
    ['Low', 'Poor', 'Temporary', 'No', 'Young', 'No'],
    ['Medium', 'Good', 'Permanent', 'Yes', 'Middle', 'Yes'],
    ['High', 'Average', 'Temporary', 'Yes', 'Old', 'No'],
    ['High', 'Good', 'Permanent', 'Yes', 'Middle', 'Yes']
]
```

```
hypothesis = data[0][:-1]
```

```
for row in data:
```

```
    if row[-1] == "Yes":
```

```
        for i in range(len(hypothesis)):
```

```
            if hypothesis[i] != row[i]:
```

```
                hypothesis[i] = "?"
```

```
print("Final Hypothesis:", hypothesis)
```

Output:

```
data = [
    ['High', 'Good', 'Permanent', 'Yes', 'Young', 'Yes'],
    ['High', 'Good', 'Permanent', 'No', 'Middle', 'Yes'],
    ['Low', 'Poor', 'Temporary', 'No', 'Young', 'No'],
    ['Medium', 'Good', 'Permanent', 'Yes', 'Middle', 'Yes'],
    ['High', 'Average', 'Temporary', 'Yes', 'Old', 'No'],
    ['High', 'Good', 'Permanent', 'Yes', 'Middle', 'Yes']
]

hypothesis = data[0][:-1]

for row in data:
    if row[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] != row[i]:
                hypothesis[i] = "?"

print("Final Hypothesis:", hypothesis)

... Final Hypothesis: ['?', 'Good', 'Permanent', '?', '?']
```

Dataset 3:

Program:

```
data = [
    ['High', 'Good', 'Yes', 'Good', 'High', 'Yes'],
    ['High', 'Excellent', 'Yes', 'Good', 'High', 'Yes'],
    ['Medium', 'Average', 'No', 'Average', 'Medium', 'No'],
    ['High', 'Good', 'Yes', 'Excellent', 'High', 'Yes'],
    ['Low', 'Poor', 'No', 'Average', 'Low', 'No'],
    ['High', 'Good', 'Yes', 'Good', 'Medium', 'Yes']
]

hypothesis = data[0][:-1]
```

```
for row in data:
```

```
    if row[-1] == "Yes":
```

```
for i in range(len(hypothesis)):
```

```
    if hypothesis[i] != row[i]:
```

```
        hypothesis[i] = "?"
```

```
print("Final Hypothesis:", hypothesis)
```

Output:

```
data = [
    ['High', 'Good', 'Yes', 'Good', 'High', 'Yes'],
    ['High', 'Excellent', 'Yes', 'Good', 'High', 'Yes'],
    ['Medium', 'Average', 'No', 'Average', 'Medium', 'No'],
    ['High', 'Good', 'Yes', 'Excellent', 'High', 'Yes'],
    ['Low', 'Poor', 'No', 'Average', 'Low', 'No'],
    ['High', 'Good', 'Yes', 'Good', 'Medium', 'Yes']
]

hypothesis = data[0][: -1]

for row in data:
    if row[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] != row[i]:
                hypothesis[i] = "?"

print("Final Hypothesis:", hypothesis)
```

```
... Final Hypothesis: ['High', '?', 'Yes', '?', '?']
```

Dataset 4:

Program:

```
data = [
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
    ['Yes', 'Yes', 'No', 'Yes', 'Yes', 'Positive'],
    ['No', 'Yes', 'Yes', 'No', 'No', 'Negative'],
    ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Positive'],
    ['No', 'No', 'Yes', 'Yes', 'No', 'Negative'],
    ['Yes', 'Yes', 'No', 'No', 'Yes', 'Positive']
]
```

```
hypothesis = data[0][: -1]
```

```
for row in data:
```

```
    if row[-1] == "Positive":
```

```
        for i in range(len(hypothesis)):
```

```
            if hypothesis[i] != row[i]:
```

```
                hypothesis[i] = "?"
```

```
print("Final Hypothesis:", hypothesis)
```

Output:

```
data = [
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
    ['Yes', 'Yes', 'No', 'Yes', 'Yes', 'Positive'],
    ['No', 'Yes', 'Yes', 'No', 'No', 'Negative'],
    ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Positive'],
    ['No', 'No', 'Yes', 'Yes', 'No', 'Negative'],
    ['Yes', 'Yes', 'No', 'No', 'Yes', 'Positive']
]

hypothesis = data[0][: -1]

for row in data:
    if row[-1] == "Positive":
        for i in range(len(hypothesis)):
            if hypothesis[i] != row[i]:
                hypothesis[i] = "?"

print("Final Hypothesis:", hypothesis)
```

```
... Final Hypothesis: ['Yes', 'Yes', '?', '?', 'Yes']
```

Dataset 5:

Program:

```
data = [
```

```
    ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Yes'],
```

```
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes'],
```

```

['No', 'No', 'No', 'No', 'No', 'No'],
['Yes', 'No', 'Yes', 'No', 'Yes', 'Yes'],
['No', 'Yes', 'No', 'Yes', 'No', 'No'],
['Yes', 'Yes', 'Yes', 'No', 'No', 'Yes']
]

```

```
hypothesis = data[0][: -1]
```

```
for row in data:
```

```
    if row[-1] == "Yes":
```

```
        for i in range(len(hypothesis)):
```

```
            if hypothesis[i] != row[i]:
```

```
                hypothesis[i] = "?"
```

```
print("Final Hypothesis:", hypothesis)
```

Output:

```

data = [
    ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'Yes'],
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes'],
    ['No', 'No', 'No', 'No', 'No', 'No'],
    ['Yes', 'No', 'Yes', 'No', 'Yes', 'Yes'],
    ['No', 'Yes', 'No', 'Yes', 'No', 'No'],
    ['Yes', 'Yes', 'Yes', 'No', 'No', 'Yes']
]

hypothesis = data[0][: -1]

for row in data:
    if row[-1] == "Yes":
        for i in range(len(hypothesis)):
            if hypothesis[i] != row[i]:
                hypothesis[i] = "?"

print("Final Hypothesis:", hypothesis)

```

... Final Hypothesis: ['Yes', '?', 'Yes', '?', '?']

Result:

The FIND-S algorithm successfully learned the most specific hypothesis for each dataset by considering only the positive training examples and ignoring the negative examples. The resulting hypotheses identify the essential attribute values required for a positive classification while replacing varying attributes with ?, indicating that those attributes do not affect the final decision.